Find

i) Equivalent emf of the combination

ii) Equivalent internal resistance

iii) Total current

iv) Potential difference across external resistance

v) Potential difference across each cell

Solution

i) Equivalent emf of the combination $\varepsilon_{eq} = n\varepsilon = 4 \times 9 = 36$ V

ii) Equivalent internal resistance $r_{eq} = nr$ $= 4 \times 0.1 = 0.4\ \Omega$

iii) Total current $I = \dfrac{n\varepsilon}{R + nr}$

$$= \frac{4 \times 9}{10 + (4 \times 0.1)}$$

$$= \frac{4 \times 9}{10 + 0.4} = \frac{36}{10.4}$$

$$I = 3.46 \text{ A}$$

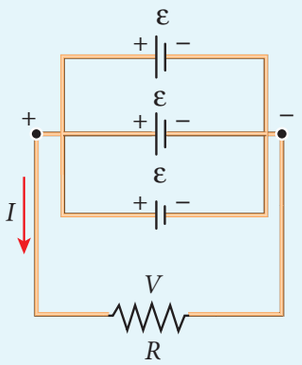
iv) Potential difference across external resistance $V = IR = 3.46 \times 10 = 34.6$ V. The remaining 1.4 V is dropped across the internal resistances of cells.

v) Potential difference across each cell $\dfrac{V}{n} = \dfrac{34.6}{4} = 8.65 \text{V}$

2.4.4 Cells in parallel

In parallel connection all the positive terminals of the cells are connected to one point and all the negative terminals to a second point. These two points form the positive and negative terminals of the battery.

Let n cells be connected in parallel between the points A and B and a resistance R is connected between the points A and B as shown in Figure 2.22. Let ε be the emf and r the internal resistance of each cell.



Cells in parallel (circuit diagram)

Figure 2.22 Cells in parallel

The equivalent internal resistance of the battery is $\dfrac{1}{r_{eq}} = \dfrac{1}{r} + \dfrac{1}{r} + ... \dfrac{1}{r} (n \text{ terms}) = \dfrac{n}{r}$. So $r_{eq} = \dfrac{r}{n}$ and the total resistance in the circuit $= R + \dfrac{r}{n}$. The total emf is the potential difference between the points A and B, which is equal to ε. The current in the circuit is given by

$$I = \frac{\varepsilon}{\frac{r}{n} + R}$$

$$I = \frac{n\varepsilon}{r + nR} \qquad (2.42)$$

Case (a) If $r >> R$, $I = \dfrac{n\varepsilon}{r} = nI_1$ (2.43)

where I_1 is the current due to a single cell $\left(\dfrac{\varepsilon}{r}\right)$ when R is negligible. Thus, the current through the external resistance due to the whole battery is n times the current due to a single cell.

Case (b) If $r << R$, $I = \dfrac{\varepsilon}{R}$ (2.44)

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